

APD background and interference

Fit comparison for discussion with Richard

6 October 2026 • Daping / Richard • Analysis of the supplied 40 s APD recording. This is a comparison of saved fits and sensitivity tests, not a calibration of absolute orientation. No new optimization was performed to prepare this report.

Main result. The inferred background and polar-angle range depend strongly on the allowed field. The earlier ~55% background solution is not uniquely required: a broader stationary-field model reaches similar agreement at 20–30%, and the tested axial-motion model reaches similar agreement at 10%. These explanations cannot yet be distinguished with this recording alone.

What to discuss first. Figures 1–2 show what the models actually fit; Figure 3 compares background makeup and angle ranges; Figure 5 shows how large background can coexist with the measured intensity through signed interference. The rest records the assumptions, controls and limits.

Model	BG / Tmax (%) 30 s / 10 s	Held-out XY RMS 30 s / 10 s	θ range (°) 30–31 s	θ range (°) 10–11 s
BG only	31.7 / 34.6	0.0896 / 0.0989	37.3–89.2	42.0–90.0
Richard effective	38.5 / 39.9	0.0927 / 0.1024	34.6–89.9	39.0–89.7
5-mode shared, relaxed	54.7 / 54.4	0.0281 / 0.0460	12.5–89.7	17.8–89.7
General stationary, $\leq 30\%$	30.0 / 29.8	0.0288 / 0.0387	30.9–64.8	32.7–63.3
General stationary, relaxed	55.0 / 54.7	0.0245 / 0.0408	6.4–89.7	11.6–89.8
Axial $R \leq 300$ nm, $\leq 10\%$	10.0 / 9.9	0.0308 / 0.0411	34.6–55.2	37.2–53.7

Direct inversion without background gives $\theta = 27.4\text{--}57.7^\circ$ at 30–31 s and $26.9\text{--}57.0^\circ$ at 10–11 s. These are descriptive baselines, not known angles. All θ ranges are folded to $0\text{--}90^\circ$ and sampled at the 128 fitted bins; none is a confidence interval.

Three conclusions supported by these calculations: (i) simply changing the measured hole outline has very little effect in the tested pupil-mask model; (ii) the restricted hole-edge field tested here is insufficient; (iii) stationary-field structure and source position can trade against background power and recovered angles. A good cone fit alone cannot resolve that tradeoff.

Stopping point. Retain $NA_{in} = 0.38$ and the current correction matrix. Do not choose a final background subtraction or publish absolute θ from the best residual alone. Decide which physical constraint or perturbation is worth measuring next.

1. Data, trajectory and fixed optical baseline

The raw file has four APD voltage channels, 250 kHz sampling and 40 s duration. Acquisition order (90°,45°,135°,0°) is explicitly reordered to (0°,90°,45°,135°). Historical gain estimates are held fixed, applied before the current inverse optical matrix, including its existing positive-beta numerical convention and /8 scale. Electronic offsets have not been independently measured for this recording; optical background fits must not be interpreted as an offset calibration.

For corrected channels, $X=(I_0-I_{90})/(I_0+I_{90})$ and $Y=(I_{45}-I_{135})/(I_{45}+I_{135})$. Every anisotropy panel uses -1 to +1 on both axes. The four channels and their total are retained in fitting; fitting the XY outline alone would discard useful power information.

The initial notebook audit found that strong periodic smoothing, spatial ridge snapping and nearest-point selection could distort an individual trajectory and mix visits to the two branches. Invalid denominators could become fake origin points; manual detection and matrix-toggle behavior also needed fixes. The replacement diagnostic averages original channel samples in contiguous, ordered bins. It does not fit a cone to construct the trajectory.

The repeatability study then uses a data-specific outer-loop crossing to detect complete observed traversals. It averages channels before forming ratios, with equal weight per traversal. The broad two-pass shape repeats, but chronological groups differ. The 10–11 s interval is more internally consistent than 30–31 s. The initial selected cycle reaching $Y \approx 0.54$ was not evidence that the ensemble average must reach that height; that interpretation was corrected.

Window	Detected traversals	Alternating train / test	Ordered bins
30–31 s	304	152 / 152	128
10–11 s	217	109 / 108	128

The saved fitting averages use contiguous ordered bins without the optional dynamic time warping. The fitted phase is assumed to advance one mechanical revolution per extracted complete trajectory; this assignment is a model assumption, not independently measured. Alternate-cycle tests measure repeatability within each window; they are not independent experiments, chronological extrapolation, or known-angle validation. Geometry, timing and brightness are fitted separately in the two windows even when the optical background is shared. Registration uncertainty and between-window drift are not propagated into formal error bars.

Collection baseline: $NA=0.38\text{--}1.30$, $n_{\text{water}}=1.33$, $n_{\text{oil}}=1.51$. Water-oil Fresnel factors and BFP amplitude mapping $\sqrt{\cos(\theta_{\text{oil}})/\cos(\theta_{\text{water}})}$ are included. The normalized scalar coefficients are $A=0.89080598$, $B=0.10919402$, $C=0.92778057$, with $A+B=1$. These A,B,C are scalar intensity coefficients, not the optical mixing matrix.

For the ideal annular rod-only model, $r=C \sin^2\theta/(A+B \sin^2\theta)$, hence $\sin^2\theta=Ar/(C-Br)$; $\phi=\frac{1}{2}\text{atan2}(Y,X)$ modulo 180°. The field fits instead infer a cone and its ordered orientations. Near-equatorial θ is poorly conditioned. Lab ϕ is not mechanical cone phase χ , and there is no universal “half a mechanical turn = one anisotropy loop” rule for a tilted cone.

2. What is being fitted — one consistent power balance

$$I_j(x) = S_j(x) + C_j(x) + B_{\text{coh},j} + B_{\text{res},j}$$

$S_j = \int |E_{s,j}|^2$, $C_j = 2 \operatorname{Re} \int E_{s,j} E_{b,j}^*$ and $B_{\text{coh},j} = \int |E_{b,j}|^2$. $B_{\text{res},j}$ is noninterfering intensity for the modeled signal. The APD integrates local intensity; it does not sum fields from unrelated pupil positions before squaring. C may be negative. It is not a positive fraction of the background, and should not be shown as a pie-chart component.

The rod field is $E_{s,j}(u,d) = a_0(d_x + i d_y) \sum_l d_l F_{jl}(u)$. Thus the excitation amplitude is $a_0 \sin \theta$, and its phase is retained. a_0 is one constant scale per interval: laser power, rod response and common collection scale are not separately calibrated. There is no independently adjustable brightness at every point in the main comparisons. The earliest free-brightness fits are retained only as historical diagnostics.

The fixed cone has three geometric parameters. A monotone mechanical-phase mapping uses 16 parameters (one starting phase and 15 relative increments) to handle variable speed. Together with a_0 , that is 20 local parameters per interval. These phase parameters move the predicted orientations along a single cone; they do not add a free interference value at each point. They still introduce model flexibility and are not independent measurements of motor phase.

Family	Additional background freedom	Sharing
Additive only	3 positive Stokes intensity parameters; $C=0$	Separate per window
Richard effective	3 Stokes + 4 bounded real channel overlaps	Separate per window
Uniform field	2 complex field coefficients + 3 Stokes = 7 real	Separate per window
Uniform + radial	3 complex coefficients + 3 Stokes = 9 real	Separate and shared tests
5-mode field	5 complex coefficients + 3 Stokes = 13 real	Shared across both windows
General stationary	10 complex coefficients + 3 Stokes = 23 real	Shared across both windows
Hole-edge surrogate	6 complex coefficients, no extra intensity = 12 real	Shared across both windows
Axial-motion model	General field + radius and phase per window	Shared BG; 4 motion parameters

Richard's operational comparison is $I_j = S_j + B_j + 2\sqrt{(S_j B_j)} c_j$, with $|c_j| \leq 1$, constant per channel. It respects individual scalar bounds but does not by itself prove a shared spatial vector field exists, or identify how much B is coherent. Its fit is therefore labeled "composition not identified."

The five-mode family contains uniform x/y fields and orthogonalized collected z-, x- and y-dipole patterns; the three-mode version contains uniform x/y plus the radial z-dipole pattern. "BG only" here is a fair forward-channel comparison with fixed excitation, not the exact historical six-parameter shape-only optimizer, whose outputs are archived separately. All fitting minimizes mean squared channel residuals divided by measured total at each bin. Reported XY RMS = $\sqrt{\text{mean}[(\Delta X)^2 + (\Delta Y)^2]}$ is a separate diagnostic. The main comparison does not drop difficult bins as background increases. Shared-field caps use the smaller training Tmax so the stated budget holds in both windows.

3. What the older model families can reproduce

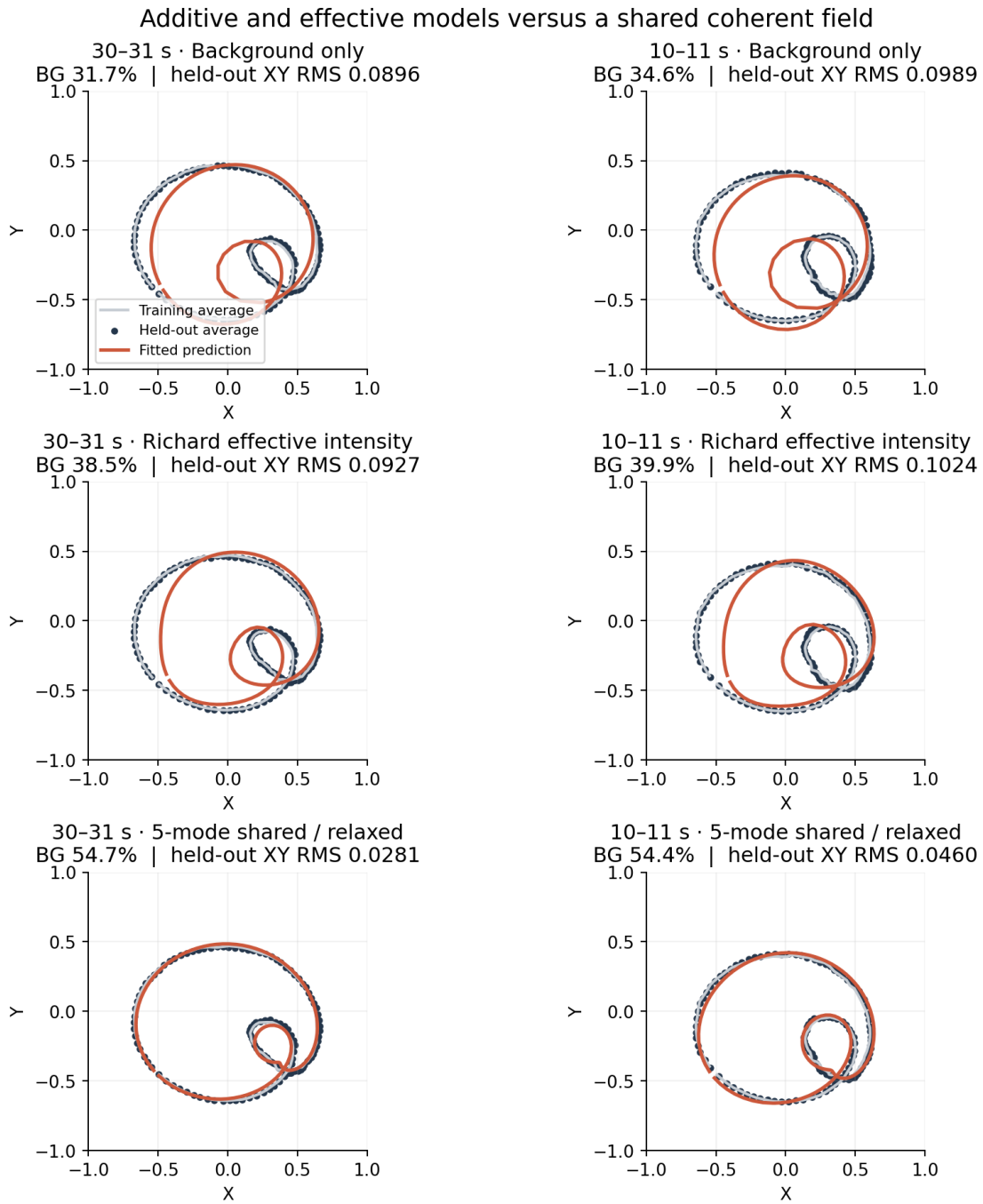


Figure 1. Points are held-out cycle averages; grey is the training average; red is the saved prediction. The additive and Richard-effective fits are separate per interval. The five-mode field uses the same absolute background coefficients in both intervals. All three use constant a_0 with circular-excitation dependence. Richard's model need not win on XY RMS because that is not the channel fitting objective.

4. Comparable fits with smaller background budgets

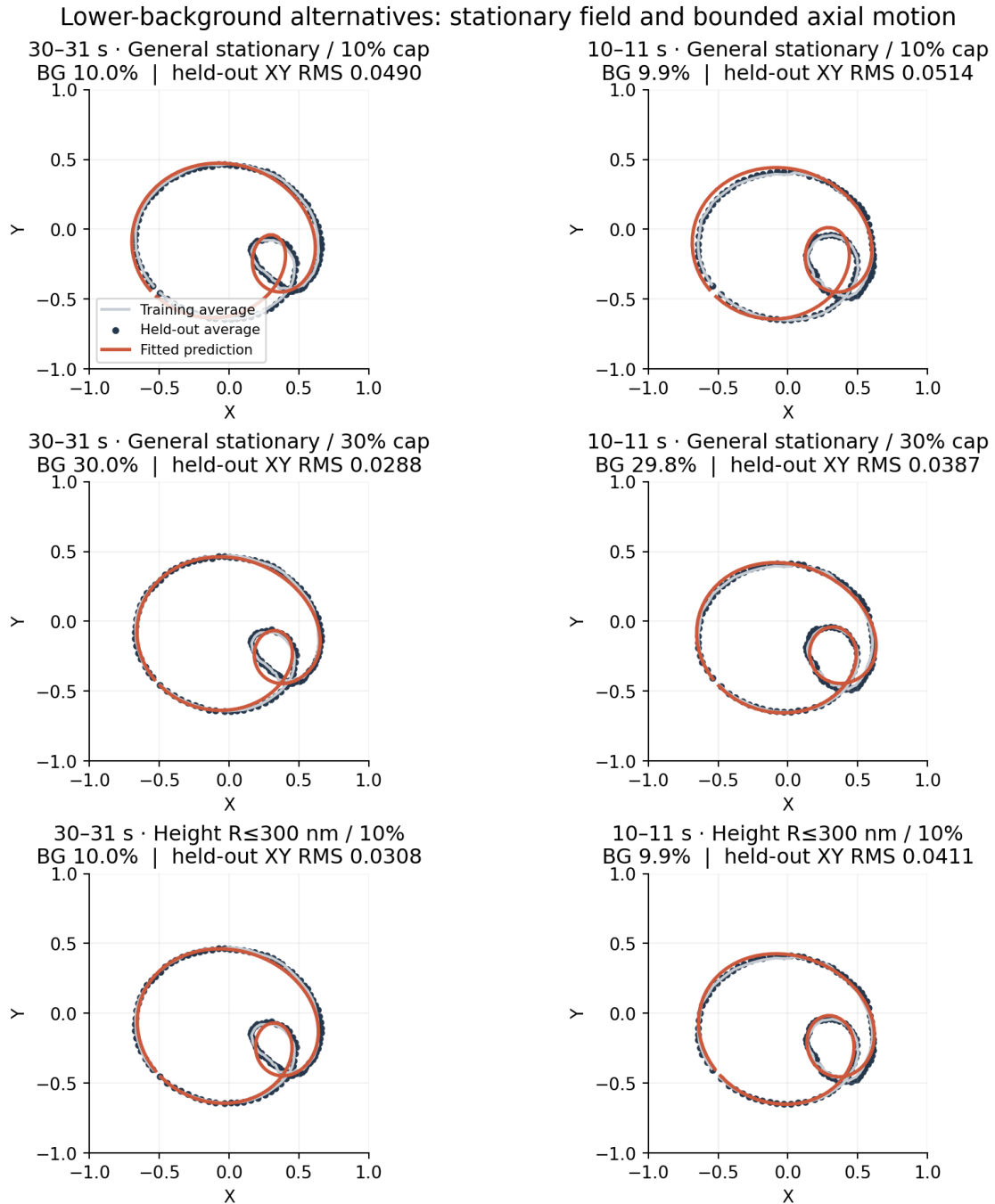


Figure 2. General stationary field at 10% and 30% caps, and the same ten-mode family with bounded axial motion at 10%. The last row is a sensitivity scenario, not evidence that the fitted motion actually occurs. These curves show why the earlier large-background estimate cannot be treated as a data-only requirement.

5. Background makeup and the angle consequences

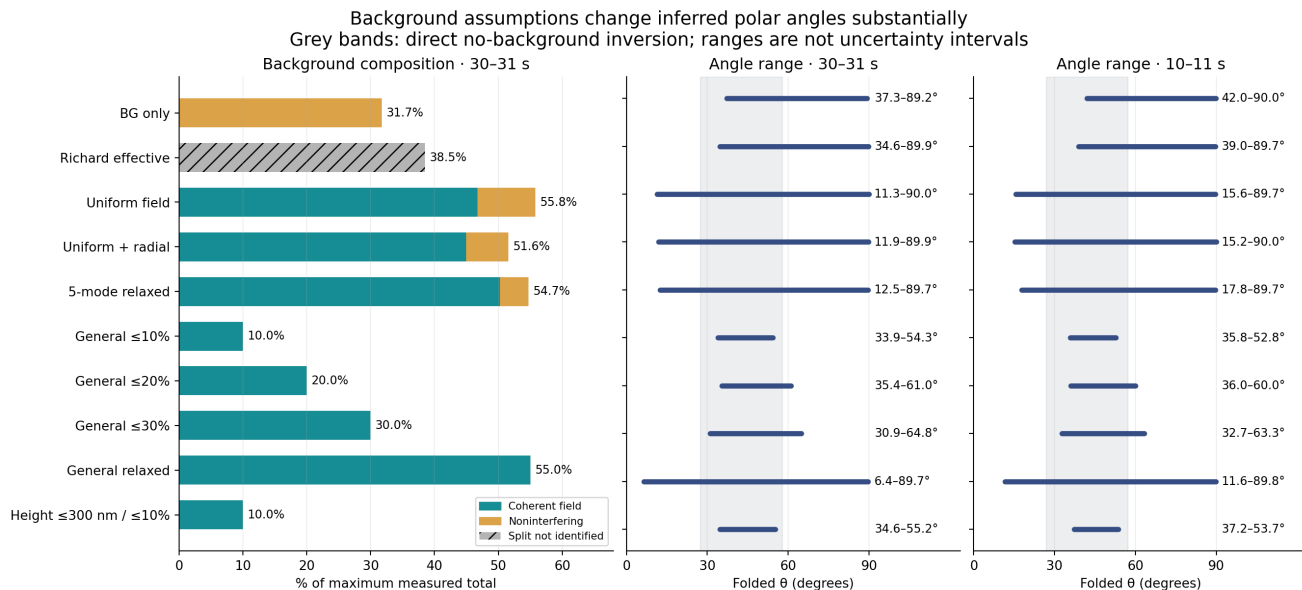


Figure 3. Background is normalized to the maximum training total, $T = \sum I_j$. The decomposition bars show 30–31 s; exact values for both windows are tabulated in the appendix. Grey angle bands show direct inversion. Changing the field family changes θ substantially, not just the residual.

The five-mode shared relaxed fit has $B/T_{\max} = 54.72\% / 54.43\%$ (30 s / 10 s). Its coherent share is 91.82%, giving coherent power 50.25% / 49.97% of $T_{\max}$ and noninterfering power 4.47% / 4.45%. In the general stationary and axial-motion selected fits, the residual Stokes component is effectively zero: essentially all fitted background is in the represented coherent field.

This split is conditional on the chosen mode space. “Noninterfering” can include light temporally incoherent with the rod, or coherent light spatially orthogonal to every signal mode being tested. Conversely, the represented coherent field can contain components with weak overlap at some orientations. Neither split assigns a percentage to coverslip, cell, objective or hole wall.

Historical “about 90–95% background” referred to the **minimum** measured total in some earlier fits. It does not mean 90–95% of the maximum. Both denominators are reported in the appendix. Nor does $B \approx T$ at a dim point imply $S \approx 0$ when C is negative: $T = S + C + B$, so destructive interference can conceal substantial rod-only power.

The unrestricted-overlap construction is deliberately excluded from these fitted-angle bars: it can reproduce the training curve exactly at a 10% background budget, but it chooses overlap independently at each point. That demonstrates feasibility under weak bounds, not a valid stationary-field correction or validated angle recovery.

6. Power decomposition: why large background can survive

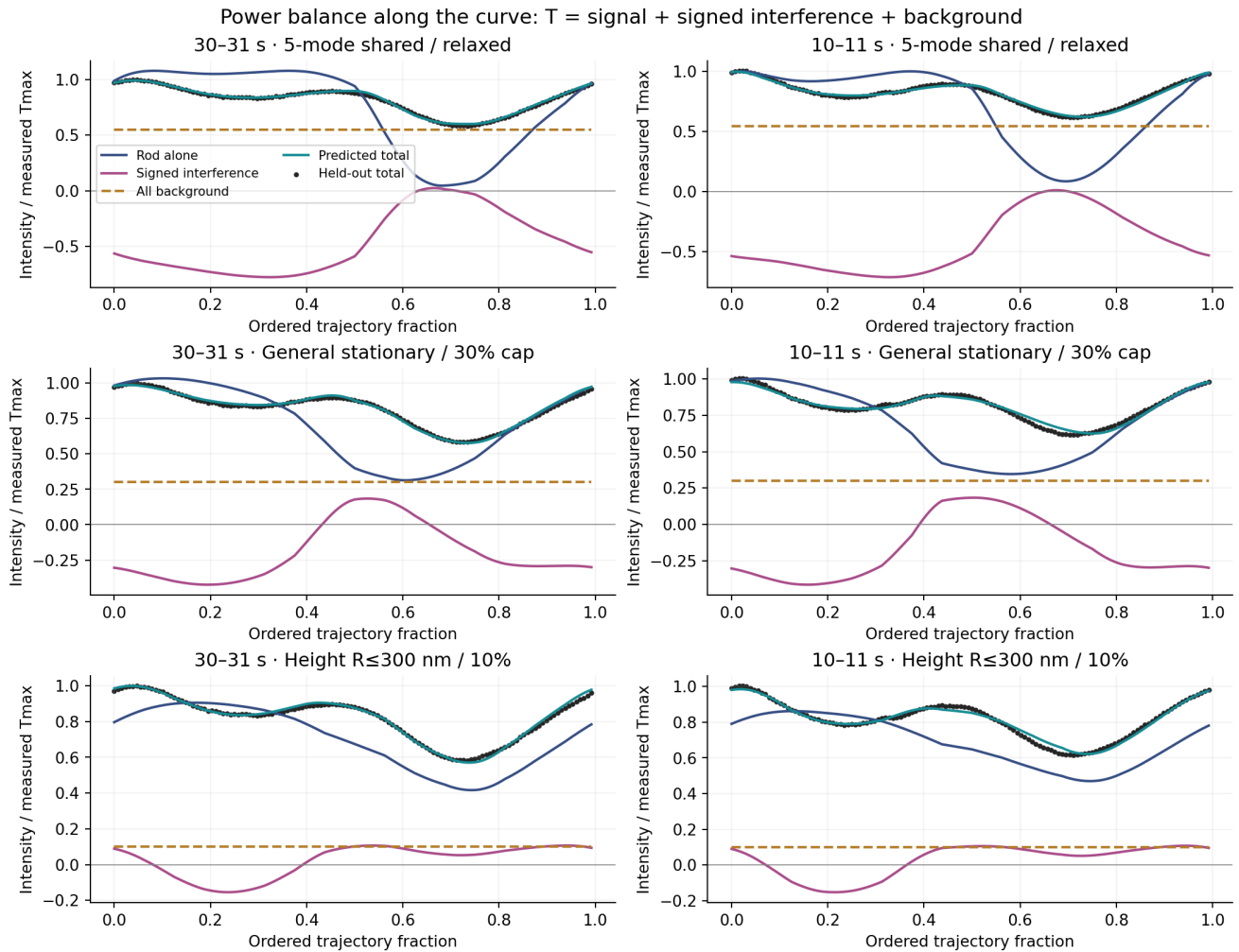


Figure 5. All terms use the same $T_{\max}$ normalization within each window. The rod-alone term obeys the one- a_0 excitation and collection law. The constant background and signed interference can compensate over substantial parts of the trajectory; the amount is not confined by an assumed positive “interference fraction.” Exact per-bin values are in comparison.json.

This checks the proposed intensity consistency condition: the recovered θ and the rod-only intensity use the same forward model and scale. Passing that internal consistency check does not establish the true θ , because cone geometry, field overlap and background were jointly chosen. The general 30% and moving-source 10% examples redistribute the same observed power differently.

7. Angle variation along the trajectory

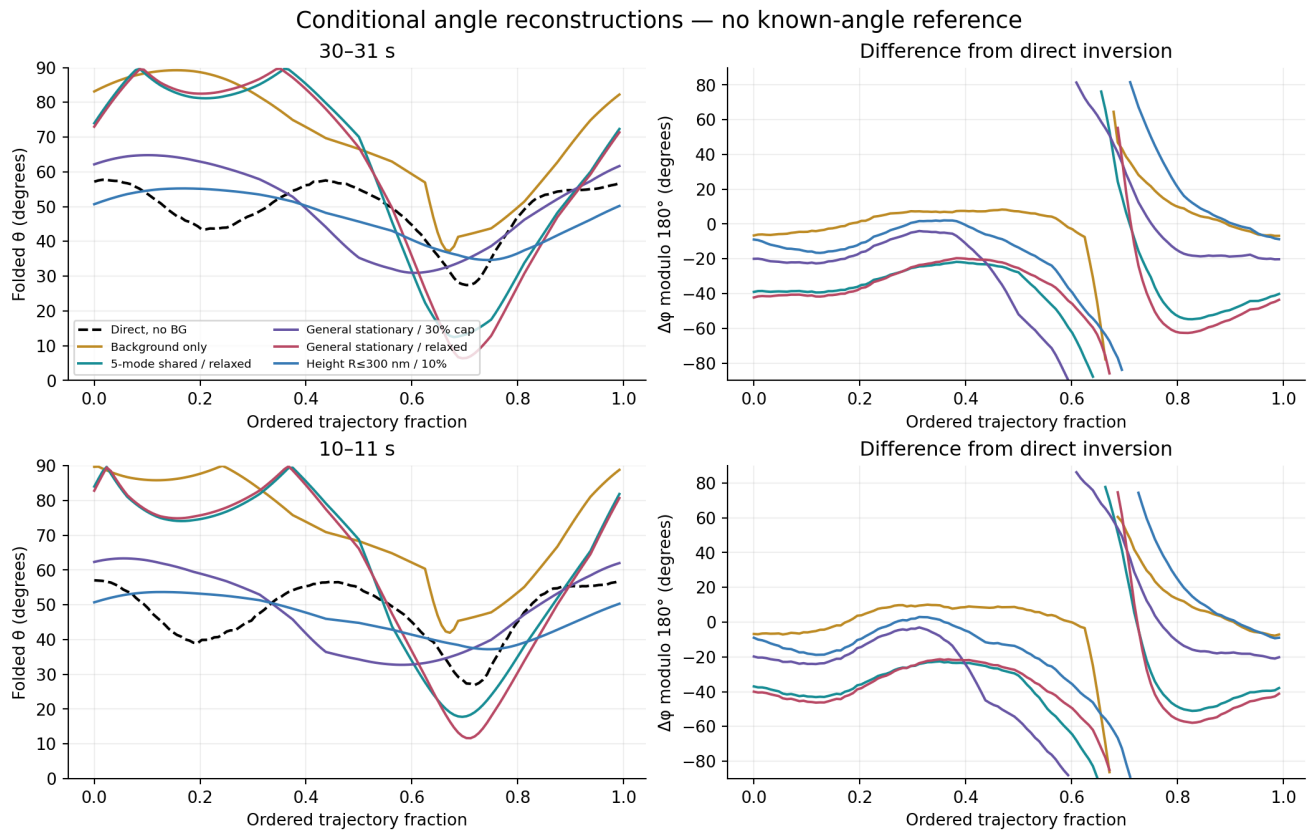


Figure 6. θ is folded to $0-90^\circ$. Azimuth differences are wrapped modulo 180° , with plot breaks at wrap discontinuities. These are comparisons at matched ordered bins, not errors relative to ground truth. Full θ and ϕ arrays, wrapped ϕ spans, and RMS differences from direct inversion are supplied in JSON/CSV.

The same measured curve admits fitted θ ranges extending almost to the equator in relaxed models, roughly $31-65^\circ$ at the 30% general stationary cap, or roughly $35-55^\circ$ in the 300 nm axial-motion scenario. Near- 90° solutions are especially sensitive to intensity/model errors. The evidence supports substantial angle ambiguity; it does not support choosing the narrowest or widest interval as more physical.

Do not interpret a wrapped ϕ minimum–maximum as a continuous rotation amplitude. The modulo- 180° trace and branch-aware difference are the appropriate comparisons. A cone imposes continuity and a branch convention but does not add an independently known orientation.

8. Stationary-field completeness and the budget tradeoff

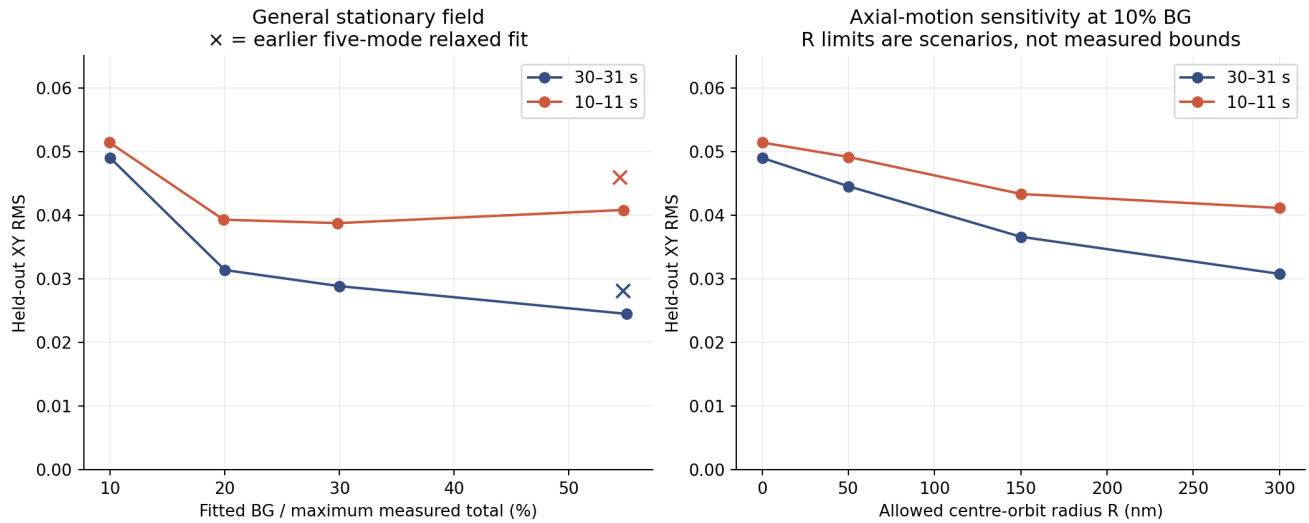


Figure 4. Left: selected general stationary fits versus their background budget, with the previous relaxed five-mode fit marked x. Right: adding bounded axial motion at 10% background. The 10-11 s held-out residual need not decrease monotonically because optimization targets training channels, not held-out XY.

For a fixed-position ideal dipole, form the scalar span V of the x/y Cartesian components of its three collected dipole fields. Its numerical rank is five. Two transverse background components in V require ten complex coefficients. The Cartesian remainder is orthogonal to V : it has no interference with any signal orientation and no integrated cross term with the represented field. Its observable residual intensity is captured by three positive Stokes parameters.

This supplies a complete observation representation for a **stationary common-pupil field under the fixed-position, ideal-dipole, full-collection assumptions**. Ten is a sufficient mode count, not a count of physical scattering objects or uniquely identifiable coefficients. Basis-dependent modal powers should not be labeled as specific sources. Adding further stationary pupil modes cannot improve this ideal fixed-position model beyond that representation.

With the circular-excitation law, the stationary rod-background cross term is quadratic in the direction d : on a fixed cone it contains mechanical-phase harmonics only through order two. Rod-only intensity is quartic and can contain order four. This restriction is why an arbitrary overlap at every bin is much more permissive than a stationary field.

The fit remains nonconvex in cone, phase and field parameters. Reported minima are feasible examples from the saved multistart searches, not proofs that a lower budget is impossible. Independent post-analyzer backgrounds, calibration errors, non-dipolar response, changing excitation and moving-source effects fall outside the completeness claim. Alternate-cycle agreement alone does not resolve them.

9. Hole shape, localized scattering and bounded height

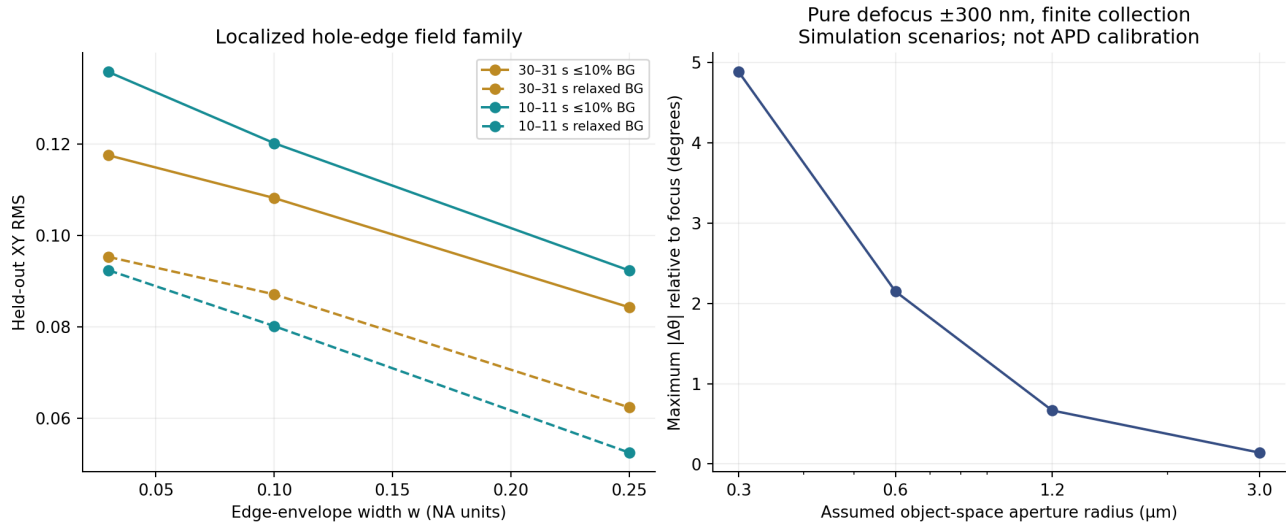


Figure 7. Left: six complex edge-field coefficients, shared across windows, with Gaussian radial envelope $\exp[-(\rho-0.38)^2/(2w^2)]$ multiplied by 1, $\cos\phi$, $\sin\phi$ in each transverse polarization. Widths 0.03, 0.10 and 0.25 NA were tested. Narrow fields fit poorly; broader fields improve but become less specifically localized to the edge. The narrow 10% selected run reached its evaluation limit; it is not a converged exclusion result.

This is an effective low-order field family, not a Maxwell calculation of the drilled wall. Wall topography, coating, mirror tilt and distance from the true BFP are not known. It cannot rule out all hole scattering. Separately, using the thesis measured amplitude mask at four rotations changed held-out XY RMS by less than about 0.0002 in the three-mode comparisons. That tests pupil transmission shape, not edge scattering or out-of-BFP propagation.

Axial-motion model: $z(\chi) = R \sin(\theta_{\text{axis}}) \cos(\chi + \delta)$, with R and δ independent per window. R is not the rod-orientation cone radius or assumed rod length. Caps 50, 150 and 300 nm are sensitivity scenarios, not biological measurements. The 300 nm fits use $R \approx 228 / 300$ nm, but axial amplitudes only 40.7 / 43.0 nm (about 81 / 86 nm peak-to-peak). Boundary-hitting radii warn against treating them as recovered hook geometry.

The rod field acquires axial phase $\exp[i(2\pi/\lambda)z(n_w + \sqrt{n_w^2 - \rho^2})]$, $\lambda = 633$ nm, including incident and collected propagation. A Gaussian illumination with 2 μm Rayleigh range and its Gouy phase is assumed. At the selected amplitudes its intensity change is below 0.05%; phase-dependent interference, rather than brightness loss, drives most of this fitted improvement.

Only axial motion is modeled; lateral motion, uncertain mean height, interface-dependent rod response and finite APD acceptance are not fitted jointly. Once the source moves, the ten-mode background is a restricted family again; fixed-position completeness no longer applies. The separate pure-defocus aperture simulation on the right tests ± 300 nm and several orientations, without background. Full collection conserves power; clipping can alter θ . These aperture radii are not calibrated APD sizes.

10. Physical interpretation and the useful stopping line

Background categories from Richard's correspondence. External optical reflections/scattering can carry arbitrary phase and spatial structure. Ideal coverslip reflection should leave through the central hole. Any reflected light that remains after bouncing or scattering need not retain a simple circular-polarization phase relation. Cell scattering can come from different heights and different polarization components; nearby rods add sample-associated fields. The APD integrates their overlap, so equal source position for the rod does not imply equal interference phase in all channels.

Multiple mutually coherent stationary backgrounds add as fields before squaring. If they lie in the same allowed mode space, fitting two copies does not add a new observation model; it only makes the decomposition redundant. Mutually incoherent contributions add intensities. More sources become observationally distinct when they add new spatial/polarization modes, different coherence, motion or a response to an experimental perturbation. Their source locations cannot generally be inferred from one steady four-channel trajectory.

Fresnel transmission, BFP mapping and the hole act on fields along the optical path; a source should enter where it is generated. After propagation to one common detection plane, the background can be represented there for this effective model. Intensity subtraction is not generally a commuting substitute for coherent propagation. The present fits do not identify which optical surface produced the effective field.

What the existing data can do: reject or expose poor versions of restricted model families; show budget-angle sensitivity; check shared-field consistency across windows; quantify registration and chronological variation; test whether physically linked terms reproduce channels as well as the outline. It cannot establish unique absolute θ , actual coherent source fractions, APD acceptance, or true centre motion without additional constraints.

Priority discussion with Richard: which perturbation best separates stationary optical field from sample-associated field in this APD setup? An axial stage scan/modulation with four-channel acquisition is especially direct, provided rod orientation, position and exposure averaging are tracked. It can constrain a phase-dependent cross term; it is not automatically an exact background subtraction.

Fixed-rod files can quantify temporal stability, channel covariance, spectra and response to laser-power steps. Flat intensity alone cannot bound a static coherent background. A nearly constant- θ rotating rod can test azimuth-dependent power and shared-field predictions, but its circular anisotropy provides less leverage for separating additive scale/background. These are useful controls, not substitutes for known-angle truth.

Other discriminating measurements are dark electronic offsets, a current transmission/mixing calibration, effective APD acceptance/defocus scans, and centre-position tracking. The sensible next step is to pick one or two constraints, not add another unconstrained per-bin term. A synthetic injection/recovery study can characterize an estimator under assumed models, but would not decide which model describes this old sample.

11. Source limits, verification and reproducible record

Bearing paper and SI. The supplied manuscript and supplementary-information (1).pdf describe deliberately selected near-axis rods with overlapping loops, fewer than 10% of candidates. Do not transfer that geometry or biological conclusions to ordinary BFM samples. SI S1 documents the drilled mirror near an inaccessible BFP, hole-wall scattering, polarization/alignment issues and APD alignment. SI p5 transmission/extinction values are historical, not a current calibration. Fixed-rod noise measurements and manual azimuth tests establish precision/ ϕ behavior, not exact θ or a coherent-background bound.

Polarcam draft and SI. Polarcam Manuscript V2 2026-09-08 RB.docx, SI outline.docx and RB Figs V2.docx were reviewed separately. Their camera background values, compensation optics, exposure/modulation behavior and some placeholder numbers are not transferred to this APD data. Axial modulation and lateral minimum projection motivate discriminating tests but require their own assumptions. APD on the Polarcam setup is not the bearing-paper APD setup.

Thesis and optical sources. Thesis sections 1.5–1.8 and its measured mirror mask supplied optical context; mask provenance is thesis commit 7576b50, analysis/image_mirror_binary.tif. The active baseline remains $NA_{in}=0.38$. Original notebook fitting history was recovered from repository import 9b3604e. The model conventions and dated source notes accompany the scripts. General pupil-phase context: <https://pmc.ncbi.nlm.nih.gov/articles/PMC5810908/> ; the present calculation uses water-side propagation with the existing water–oil transmission.

Numerical checks. This report reconstructs and checks 46 model/window predictions against saved residuals and checks S+C+B consistency. Earlier checks reproduce all 15 saved edge/general/height comparisons; arbitrary fixed-field overlap projection error is about 2.9×10^{-16} ; pupil refinement changes channels by about 1.2×10^{-4} for general fields and 7.2×10^{-4} for the narrow edge family; checked axial interpolation error is about 5.3×10^{-7} . These are numerical checks, not experimental accuracy bounds.

The best narrow-edge 10% run has success=false at 600 function evaluations. Other selected runs in the principal table meet their optimizer termination criteria. Several caps/radius constraints are active, and multistart optimization does not certify global minima. No information criterion or formal confidence interval is claimed: correlated bins, estimated registration and unknown error covariance make naive parameter-count significance unreliable.

Reproduction. comparison.csv is a compact numerical table; comparison.json includes per-bin predictions, signal/interference/background decomposition, training/test arrays and angles. build_data.py reconstructs results; plot_figures.py draws seven PNG/SVG figure sets; build_report.py creates this PDF and editable Word document. The study archive preserves earlier scripts, fit parameters, cached averages, optical inputs and dated notes. Replotting cached results does not require the large raw TDMS. Reprocessing and refitting require their documented data paths and dependencies.

No future Codex handoff script is included yet. The code and dated record are the current milestone; the report is intended to settle the scientific next step before extending the model.

Appendix A · 30-31 s: fit quality and angles

Model	XY RMS	Channel RMS %	Total RMS %	θ span (°)	ϕ RMS vs direct (°)	Param.
Background only	0.0896	2.29	5.81	37.3–89.2	17.5	23 L
Richard effective intensity	0.0927	2.14	4.80	34.6–89.9	18.2	27 L
Uniform field (2 modes)	0.0538	1.05	2.25	11.3–90.0	41.4	27 L
Uniform + radial (3 modes)	0.0317	0.60	0.98	11.9–89.9	39.7	29 L
3-mode shared / relaxed	0.0336	0.65	1.26	11.1–89.8	40.3	49 S
3-mode shared / 10% cap	0.0950	2.15	4.88	26.6–53.9	19.3	49 S
5-mode shared / relaxed	0.0281	0.59	1.15	12.5–89.7	41.5	53 S
5-mode shared / 10% cap	0.0930	2.11	4.75	26.4–53.8	19.5	53 S
Edge $w=0.03$ / 10% cap	0.1175	2.66	5.97	36.2–57.3	16.8	52 S *
Edge $w=0.03$ / relaxed	0.0953	2.03	4.63	33.7–88.6	19.5	52 S
Edge $w=0.1$ / 10% cap	0.1082	2.45	5.53	34.5–56.5	18.7	52 S
Edge $w=0.1$ / relaxed	0.0871	1.87	4.39	35.8–79.6	21.3	52 S
Edge $w=0.25$ / 10% cap	0.0843	1.93	4.50	33.1–55.4	20.0	52 S
Edge $w=0.25$ / relaxed	0.0624	1.36	3.23	36.7–66.7	25.8	52 S
General stationary / 10% cap	0.0490	1.06	2.74	33.9–54.3	26.0	63 S
General stationary / relaxed	0.0245	0.49	0.87	6.4–89.7	42.6	63 S
General stationary / 20% cap	0.0314	0.67	1.52	35.4–61.0	31.6	63 S
General stationary / 30% cap	0.0288	0.62	1.36	30.9–64.8	34.2	63 S
Height $R \leq 50$ nm / 10%	0.0445	0.94	2.40	34.3–54.8	26.1	67 S
Height $R \leq 150$ nm / 10%	0.0366	0.75	1.80	33.8–56.2	27.0	67 S
Height $R \leq 300$ nm / 10%	0.0308	0.64	1.39	34.6–55.2	28.5	67 S

Channel and total RMS use residual divided by observed total at each bin; channel RMS is not relative to each channel individually. θ span covers the sampled fitted orientations. ϕ differences use modulo 180°. S = total parameters in the joint two-window fit; L = parameters in this separate single-window fit. * = selected run reached its evaluation limit. Numeric termination does not establish global optimality.

Direct no-background inversion is not a fitted cone and has no fitted prediction score here. All historical and verified duplicates are retained in comparison.json/CSV. Early free-brightness and arbitrary-overlap constructions are not comparable predictive models and are intentionally kept outside this table.

Appendix B · 30-31 s: background composition

Model	All BG % Tmax	All BG % Tmin	Coherent % Tmax	Residual % Tmax	Channel BG % Tmax 0 / 90 / 45 / 135
Background only	31.74	54.07	0.0	31.7	9.3/6.6/4.7/11.1
Richard effective intensity	38.51	65.60	—	—	13.5/5.7/9.1/10.2
Uniform field (2 modes)	55.77	95.00	46.8	9.0	15.4/12.5/10.4/17.5
Uniform + radial (3 modes)	51.57	87.85	44.9	6.6	16.1/9.7/9.7/16.0
3-mode shared / relaxed	53.05	90.37	46.2	6.9	16.8/9.7/9.9/16.7
3-mode shared / 10% cap	10.00	17.04	4.3	5.7	4.2/0.8/1.3/3.7
5-mode shared / relaxed	54.72	93.23	50.3	4.5	17.9/9.4/10.6/16.7
5-mode shared / 10% cap	10.00	17.04	5.5	4.5	4.1/0.9/1.5/3.5
Edge w=0.03 / 10% cap	10.00	17.04	10.0	0.0	3.5/1.5/0.4/4.6
Edge w=0.03 / relaxed	34.90	59.46	34.9	0.0	11.5/5.9/6.7/10.8
Edge w=0.1 / 10% cap	10.00	17.04	10.0	0.0	4.1/0.9/1.0/4.0
Edge w=0.1 / relaxed	31.90	54.35	31.9	0.0	10.9/5.1/5.8/10.2
Edge w=0.25 / 10% cap	10.00	17.04	10.0	0.0	4.3/0.7/1.3/3.7
Edge w=0.25 / relaxed	25.63	43.66	25.6	0.0	9.8/3.1/4.0/8.8
General stationary / 10% cap	10.00	17.04	10.0	0.0	4.1/0.9/1.3/3.7
General stationary / relaxed	55.04	93.76	55.0	0.0	17.3/10.3/11.3/16.3
General stationary / 20% cap	20.00	34.07	20.0	0.0	9.0/1.0/3.0/7.0
General stationary / 30% cap	30.00	51.11	30.0	0.0	12.2/2.8/4.3/10.7
Height $R \leq 50$ nm / 10%	10.00	17.04	10.0	0.0	4.1/0.9/1.3/3.7
Height $R \leq 150$ nm / 10%	10.00	17.04	10.0	0.0	4.0/1.0/1.0/4.0
Height $R \leq 300$ nm / 10%	10.00	17.04	10.0	0.0	3.8/1.2/0.9/4.1

Tmax and Tmin are from the training total within this window. Channel entries sum to all BG up to rounding. Coherent/residual components are nonnegative fitted powers; interference is signed and reported separately in Figure 5 and comparison.json. A dash means the effective model does not identify that split. Near-zero residual components are rounded to 0.0%; this is a model-dependent boundary solution, not proof that all real background is coherent.

Shared-field coefficients are identical in absolute units between windows. Their percentages differ because the observed maxima/minima differ. The background is constant along each fitted trajectory; it is the signal-background cross term that varies with orientation and, in the height scenario, position.

Appendix A · 10-11 s: fit quality and angles

Model	XY RMS	Channel RMS %	Total RMS %	θ span (°)	ϕ RMS vs direct (°)	Param.
Background only	0.0989	2.56	6.66	42.0–90.0	19.5	23 L
Richard effective intensity	0.1024	2.41	4.95	39.0–89.7	18.2	27 L
Uniform field (2 modes)	0.0709	1.30	1.69	15.6–89.7	41.8	27 L
Uniform + radial (3 modes)	0.0495	1.01	1.76	15.2–90.0	42.7	29 L
3-mode shared / relaxed	0.0509	1.04	1.69	16.1–89.9	41.7	49 S
3-mode shared / 10% cap	0.1040	2.36	4.86	29.2–52.1	18.9	49 S
5-mode shared / relaxed	0.0460	0.98	1.68	17.8–89.7	42.4	53 S
5-mode shared / 10% cap	0.1019	2.32	4.78	29.0–52.0	19.1	53 S
Edge $w=0.03$ / 10% cap	0.1357	3.14	6.60	37.9–54.2	17.7	52 S *
Edge $w=0.03$ / relaxed	0.0924	2.17	4.76	39.5–90.0	18.7	52 S
Edge $w=0.1$ / 10% cap	0.1202	2.79	5.76	36.9–53.7	17.9	52 S
Edge $w=0.1$ / relaxed	0.0802	1.97	4.62	41.8–78.6	20.9	52 S
Edge $w=0.25$ / 10% cap	0.0923	2.19	4.72	35.5–52.9	19.3	52 S
Edge $w=0.25$ / relaxed	0.0525	1.48	3.96	40.4–64.1	25.7	52 S
General stationary / 10% cap	0.0514	1.33	3.62	35.8–52.8	26.0	63 S
General stationary / relaxed	0.0408	0.91	1.92	11.6–89.8	43.1	63 S
General stationary / 20% cap	0.0393	1.03	2.65	36.0–60.0	34.7	63 S
General stationary / 30% cap	0.0387	0.99	2.47	32.7–63.3	38.3	63 S
Height $R \leq 50$ nm / 10%	0.0491	1.24	3.24	36.4–53.4	26.1	67 S
Height $R \leq 150$ nm / 10%	0.0433	1.10	2.76	36.4–54.6	27.1	67 S
Height $R \leq 300$ nm / 10%	0.0411	0.99	2.19	37.2–53.7	27.9	67 S

Channel and total RMS use residual divided by observed total at each bin; channel RMS is not relative to each channel individually. θ span covers the sampled fitted orientations. ϕ differences use modulo 180°. S = total parameters in the joint two-window fit; L = parameters in this separate single-window fit. * = selected run reached its evaluation limit. Numeric termination does not establish global optimality.

Direct no-background inversion is not a fitted cone and has no fitted prediction score here. All historical and verified duplicates are retained in comparison.json/CSV. Early free-brightness and arbitrary-overlap constructions are not comparable predictive models and are intentionally kept outside this table.

Appendix B · 10-11 s: background composition

Model	All BG % Tmax	All BG % Tmin	Coherent % Tmax	Residual % Tmax	Channel BG % Tmax 0 / 90 / 45 / 135
Background only	34.62	55.81	0.0	34.6	9.8/7.5/4.7/12.6
Richard effective intensity	39.93	64.38	—	—	14.0/6.0/9.8/10.2
Uniform field (2 modes)	58.92	95.00	45.6	13.3	17.1/12.3/10.5/19.0
Uniform + radial (3 modes)	55.34	89.23	47.9	7.4	17.6/10.1/10.1/17.6
3-mode shared / relaxed	52.76	85.07	45.9	6.8	16.7/9.7/9.8/16.6
3-mode shared / 10% cap	9.95	16.04	4.3	5.7	4.2/0.8/1.3/3.6
5-mode shared / relaxed	54.43	87.75	50.0	4.4	17.8/9.4/10.6/16.7
5-mode shared / 10% cap	9.95	16.04	5.5	4.5	4.1/0.9/1.5/3.5
Edge w=0.03 / 10% cap	9.95	16.04	9.9	0.0	3.4/1.5/0.4/4.6
Edge w=0.03 / relaxed	34.71	55.97	34.7	0.0	11.5/5.9/6.6/10.7
Edge w=0.1 / 10% cap	9.95	16.04	9.9	0.0	4.1/0.9/1.0/3.9
Edge w=0.1 / relaxed	31.73	51.16	31.7	0.0	10.8/5.0/5.7/10.1
Edge w=0.25 / 10% cap	9.95	16.04	9.9	0.0	4.3/0.7/1.3/3.7
Edge w=0.25 / relaxed	25.49	41.10	25.5	0.0	9.7/3.0/3.9/8.8
General stationary / 10% cap	9.95	16.04	9.9	0.0	4.1/0.9/1.2/3.7
General stationary / relaxed	54.73	88.25	54.7	0.0	17.2/10.2/11.2/16.2
General stationary / 20% cap	19.89	32.07	19.9	0.0	9.0/0.9/3.0/6.9
General stationary / 30% cap	29.84	48.11	29.8	0.0	12.2/2.8/4.3/10.6
Height $R \leq 50$ nm / 10%	9.95	16.04	9.9	0.0	4.1/0.9/1.3/3.7
Height $R \leq 150$ nm / 10%	9.95	16.04	9.9	0.0	4.0/1.0/1.0/4.0
Height $R \leq 300$ nm / 10%	9.95	16.04	9.9	0.0	3.8/1.2/0.9/4.0

Tmax and Tmin are from the training total within this window. Channel entries sum to all BG up to rounding. Coherent/residual components are nonnegative fitted powers; interference is signed and reported separately in Figure 5 and comparison.json. A dash means the effective model does not identify that split. Near-zero residual components are rounded to 0.0%; this is a model-dependent boundary solution, not proof that all real background is coherent.

Shared-field coefficients are identical in absolute units between windows. Their percentages differ because the observed maxima/minima differ. The background is constant along each fitted trajectory; it is the signal-background cross term that varies with orientation and, in the height scenario, position.